

SCALE-UP PLAN · 2026–2029

From one canteen to every tray.

How PlateLoop grows from a working prototype to hundreds of canteens: prove it in one school, scale through caterers, then expand into hospitals and offices.

The product works. Now we need real canteens.

BUILT

A full working prototype

- ✓ 3D scanner with its kiosk
- ✓ Kitchen dashboard and three Loopi apps
- ✓ Live demo, deck, video and poster

PROVEN ELSEWHERE

The method is backed by evidence

- ✓ **36%** less food waste when caterers measure it
- ✓ **~89%** of foods identified from tray scans
- ✓ **6:1** return on waste reduction

MISSING

Data from a real canteen

Our savings figures are estimates. A pilot turns them into measured results that caterers, hospitals and funders will trust.

THE OPPORTUNITY

Hundreds of canteens, new rules, and money to help.

317

MOE primary and secondary schools, each with a canteen

MOE, 2024: 177 primary, ~140 secondary

2024→25

Caterers and large food premises must now separate food waste

NEA, Resource Sustainability Act

80%

of a waste-reduction project's costs can be co-funded, with food waste prioritised

NEA 3R Fund (cap S\$1M per project)

755,000_t

of food wasted in Singapore in a year, only 18% recycled

NEA, 2023

Four phases. Each one earns the next.

1

NEXT TERM

Prove it

- One school canteen
- Two scanners
- Measure the real savings

1 site



2

YEAR 1

Partner

- One contract caterer
- Its school canteens
- Multi-site dashboard

10 sites



3

YEAR 2

Expand

- More caterers and schools
- First hospital ward
- First office canteen

50 sites



4

YEAR 3+

Scale

- Universities and hospitals
- Region: Malaysia, Korea, Japan
- Personal app for everyone

150+ sites

A pilot designed to prove one thing: it saves food.

How it runs

- ✓ **1 school canteen**, about 800 lunches a day
- ✓ **2 scanners**, so the queue keeps moving
- ✓ **Weeks 1–2: baseline.** Scan only, no feedback, to measure today's waste
- ✓ **Weeks 3–10: full PlateLoop.** Kitchen forecasts, Loopi and class races on
- ✓ **Consent first:** students and parents opt in; no photos leave the scanner

WE MEASURE	SUCCESS MEANS
Plate waste kg left per tray, vs the baseline	–20% or more
Food cooked kg prepared vs the old plan	–10% or more
Scan rate trays scanned before and after	90%+
Speed time per scan	≤ 3 s
Accuracy dishes correctly recognised	85%+
Liked it students and canteen staff	4 / 5

One caterer partnership opens many canteens.

Why caterers first

- ✓ **One contract, many sites:** a caterer runs canteens in many schools
- ✓ **They already have to act:** food waste segregation is now required for large food premises
- ✓ **They keep the savings:** ingredients are their biggest cost, and measuring waste returns about 6:1

What we give them

- ✓ **Scanners on lease,** nothing to buy upfront
- ✓ **One dashboard for every site:** waste, orders and nutrition side by side
- ✓ **Carbon and waste reports** ready for regulators and ESG reporting
- ✓ **Loopi for each school,** so students help cut waste too

Same scanner. Three new places to save food.

HOSPITALS

Loopi Care on one ward

Run with the ward's dietitians. Measure whether tracking every tray catches poor eaters earlier: 29% of inpatients in one Singapore study were malnourished.

OFFICES

Loopi Work in a corporate canteen

Pre-orders let the kitchen cook to real demand, and workers get a daily pick for their goal. Private by default.

UNIVERSITIES

Loopi for campuses

Large dining halls with thousands of meals a day, and students who eat too little fruit and veg: only 13.6% meet the recommendation.

Built to roll out, not just to demo.

HARDWARE**Off-the-shelf parts**

Depth camera, load cell, tablet and an edge AI module. A contract manufacturer builds the casing.

NEW MENUS**Learns new dishes fast**

Each canteen's menu is known in advance, so the AI only chooses between today's dishes. New dishes are added from labelled trays.

CLOUD**One platform, many sites**

Every scanner syncs grams per dish to one cloud, so a caterer sees all its canteens together. Works offline too.

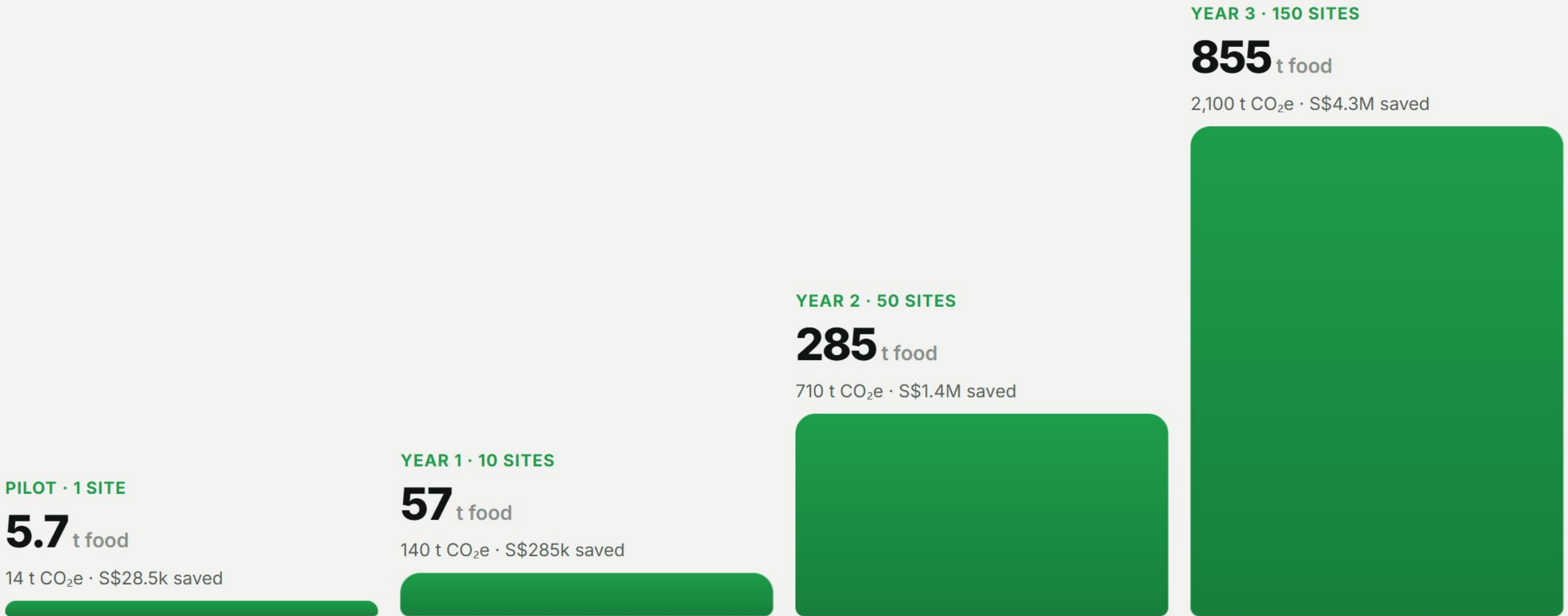
PRIVACY**PDPA by design**

No photos leave the scanner. Faces become a match code. Data is deleted on schedule, and consent comes first.

The customer saves more than it pays, from month one.

Customer saves on ingredients	\$S\$2,375 / mo	Scanner hardware (est.)	~\$S\$3,000
Customer pays (Pro plan)	\$S\$470 / mo	Our cost to run a site (cloud + support, est.)	~\$S\$80 / mo
Return for the customer	5x	We recover the hardware in	~8 months
Customer pays back a year in	2.4 months	Basic plan, no forecasts	\$S\$200 / mo

Every site we add saves about 5.7 tonnes of food.



Same assumptions as one 800-meal school (5.7 t food, 14 t CO₂e, S\$28,500 a year), multiplied by sites. Real sites vary; the pilot sets the true number.

What it takes to get there.

Funding

- ✓ **Pilot (~S\$10k):** two scanners, cloud and installation. Apply to NEA's 3R Fund, which co-funds up to 80%
- ✓ **Year 1:** lease income from the caterer, plus innovation grants
- ✓ **Year 2:** a seed round to build scanners in volume and enter hospitals

Team

- ✓ **Today:** our student team, who designed and built the prototype
- ✓ **Next:** a hardware engineer and a machine-learning engineer
- ✓ **Then:** a partnerships lead for caterers, and a dietitian adviser for hospitals
- ✓ **Mentors:** a canteen operator and a school sustainability lead

RISKS

What could go wrong, and what we'll do.

RISK	HOW WE HANDLE IT
Mixed dishes are hard to recognise	Compartment trays, a known daily menu, and the weighing platform to check every estimate.
Students forget to scan after lunch	Loopi only eats if you scan; reminders on the scanner; the kitchen sees trays still out.
Queues get slower	About 2 seconds a scan, and one scanner for roughly every 400 diners.
Privacy concerns	No photos leave the scanner, consent on day one, face sign-in can be switched off.
Savings smaller than expected	The pilot measures them first; pricing stays below the measured saving.

OUR ASK

One school canteen. One term.

A canteen

to host two scanners for ten weeks

A partner

to co-apply for NEA's 3R Fund with us

An introduction

to a contract caterer for Phase 2

We'll bring the scanners and the apps, and share every number with the school and the canteen. **PlateLoop: know what's eaten. Waste less.**